7 **(a)** By reference to the photoelectric effect, state what is meant by the *threshold frequency*.

.....

..... [1]

(b) Electrons are emitted from a metal surface when light of a particular wavelength is incident on the surface. Explain why the emitted electrons have a range of values of kinetic energy below a maximum value.

.....

.....

.....

..... [2]

(c) The wavelength of the incident radiation is λ. The variation with $1/\lambda$ of the maximum kinetic energy E_{MAX} of electrons emitted from a metal surface is shown in Fig. 7.1.

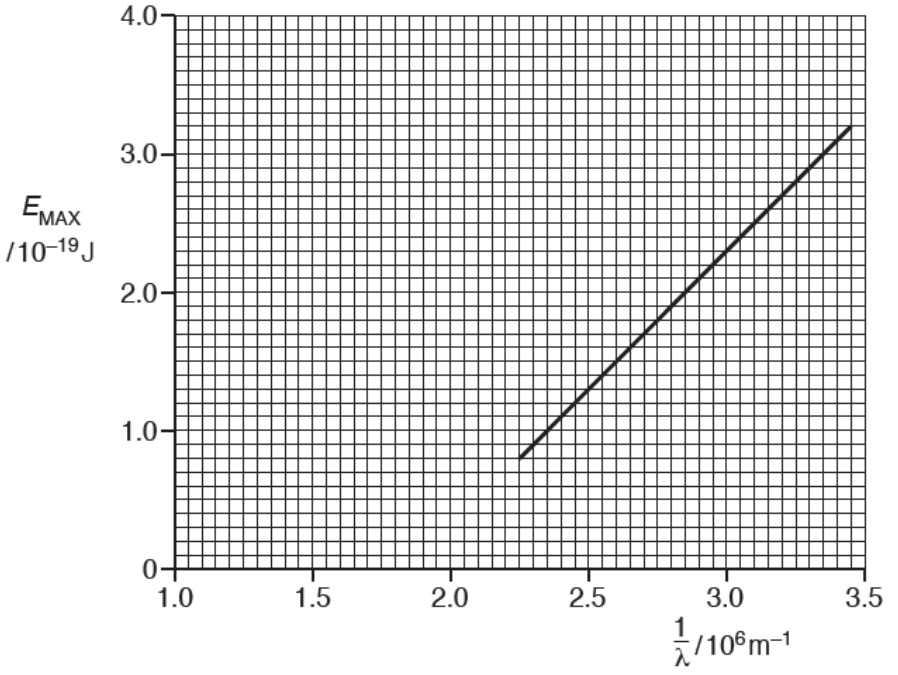

Fig. 7.1

(i) Use Fig. 7.1 to determine the threshold frequency f_0.

f_0 = ... Hz [2]

(ii) Use your answer in **(i)** to calculate the work function energy Φ.

Φ = ... J [2]

(iii) State an evidence from Fig. 7.1 that the classical wave theory cannot explain.

.....

.....

..... [1]

(iv) If the intensity of the radiation is doubled, state the changes in the graph, if any.

.....

.....

..... [1]

(v) Caesium metal has a lower work function energy than the metal in **(c)**.

On the axes of Fig. 7.1, sketch a line to show the variation with $1/\lambda$ of E_{MAX} for caesium metal. Label this line **C**.

[2]